

MODEL TRAINING REPORT

Electronic Nose Coffee Classification System

Model: Convolutional Neural Network (1D-CNN)

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TRAINING SUMMARY

Training Accuracy	Validation Accuracy	Training Time
97.4%	93.8%	3.0min

MODEL CONFIGURATION

Model Type	1D Convolutional Neural Network (1D-CNN)
Epochs	75
Data Split	80% Training / 20% Validation
Input Shape	8 sensors x 300 timesteps
Architecture	1D-CNN + Backpropagation (SGD, Cross Entropy)

GAS SENSORS USED (8 Sensors)

TGS2600	MQ135	MQ3	MQ6
MQ7	TGS2602	TGS2611	TGS2620

POWER CONSUMPTION DURING TRAINING

Power Metric	Value	Thermal / Energy	Value
Avg Total Power	6.47 W	Avg CPU Temp	47.4 °C
Peak Total Power	7.06 W	Peak CPU Temp	48.0 °C
Avg CPU+GPU Power	1.90 W	Energy Used	1114.9 J (0.3097 Wh)
Peak CPU+GPU Power	2.31 W	Training Duration	3.0 min

MODEL EVALUATION RESULTS

Training Data — Accuracy: 97.40%

Metric	High Grade (Arabica)	Low Grade (Robusta)
Precision	0.9691	0.9789
Recall	0.9792	0.9688
F1-Score	0.9741	0.9738
Support (n samples)	96	96

Validation Data — Accuracy: 93.75%

Metric	High Grade (Arabica)	Low Grade (Robusta)
Precision	0.9565	0.9200
Recall	0.9167	0.9583
F1-Score	0.9362	0.9388
Support (n samples)	24	24

Confusion Matrix (Validation Data)

	Predicted: High	Predicted: Low
Actual: High	22	2
Actual: Low	1	23

TP = True Positive (High predicted as High)

FP = False Positive (Low predicted as High)

TN = True Negative (Low predicted as Low)

FN = False Negative (High predicted as Low)

MODEL DETAILS & TRAINING CURVES

1D-CNN Summary -- 75 Epochs

Train Loss	0.074607
Validation Loss	0.156083
Total Epochs	75

Accuracy Curve (Checkpoint)

Epoch	Training Accuracy (%)	Validation Accuracy (%)
1	50.00	50.00
9	57.81	50.00
17	92.71	87.50
25	94.79	89.58
33	96.88	91.67
42	97.40	91.67
50	96.88	93.75
58	96.88	93.75
66	97.40	93.75
75	97.40	93.75

Loss Curve (Checkpoint)

Epoch	Training Loss	Validation Loss
1	0.693915	0.692998
9	0.690374	0.689046
17	0.315947	0.343572
25	0.123800	0.196795
33	0.090777	0.200908
42	0.085557	0.202918
50	0.083183	0.192935
58	0.080553	0.179987
66	0.077547	0.167258
75	0.074607	0.156083